

SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 1.4: Energy, Work, Power and Machines — Teacher Reference (Pre-filled)

SUMMARY TABLE: PHYSICS GRADE 10	
Sub-Strand	Sub-Strand 1.4: Energy, Work, Power and Machines
Driving Question	DRIVING QUESTION: How do pulleys reduce the effort needed to lift water, and how does energy transfer in the process?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Kinetic and Potential Energy: The Bucket in the Well	In the Energy Skate Park simulation we observed that the skater has the greatest potential energy at the highest point and the greatest kinetic energy at the lowest point, with energy moving between the two as the skater moves. We also observed from the bucket table that the higher the bucket is in the Kitui well, the more potential energy it has, rising from 0 J at the water to 1600 J at the top.	Potential energy is energy stored because of position, such as a bucket lifted up the well, and kinetic energy is energy an object has because it is moving. A bucket lifted higher stores more potential energy, and stored energy can change into movement energy when something falls.	We can explain that Mama Kavata and Mwende both raise the bucket the same 8 m, so the bucket gains stored (potential) energy as it rises in both cases. We cannot yet explain why the pulley makes it easier, where the energy for lifting comes from, or whether any energy is lost; these questions are on our Driving Question Board for the lessons ahead.
Lesson 2 Energy Transformations and the Formulas $KE = \frac{1}{2}mv^2$ and $PE = mgh$	We swung a 0.5 kg pendulum bob raised 0.2 m and watched it move fastest at the bottom and stop at the top of its swing. We saw a roller coaster car speed up going down a hill and slow down going up. We read the free-fall table for the 20 kg bucket from 8 m to 0 m, where PE falls from 1600 J to 0 J as KE rises from 0 J to 1600 J.	$PE = mgh$ and $KE = \frac{1}{2}mv^2$, with $g = 10 \text{ N/kg}$ and energy in joules. The pendulum has 1 J of PE at the top and 1 J of KE (2 m/s) at the bottom. The roller coaster car has 200 000 J of PE at the top and 200 000 J of KE (20 m/s) at the bottom in the ideal case. Energy changes form between PE and KE. The bucket gains 400, 800, 1200 and 1600 J of PE at 2, 4, 6 and 8 m.	The bucket in the Kitui well gains 1600 J of PE when raised 8 m ($200 \text{ N} \times 8 \text{ m}$), whether Mama Kavata lifts it by hand or Mwende uses the pulley system, because PE depends only on weight and height. This does not yet explain why the pulley feels easier or where the energy comes from; those questions stay on the DQB for the next lessons.
Lesson 3 Conservation of Energy: Does the Total Stay the Same? Pendulum and the Falling Bucket	In the PhET Pendulum Lab and in the free-fall table for the 20 kg bucket released from 8 m, we observed that PE decreases and KE increases as the object moves down, while the total $KE + PE$ stays the same: 1600 J at every height for the bucket and 1	We learned that in an ideal system (no friction) the total energy stays constant. Energy is never created or destroyed, only transformed (PE to KE and back) or transferred. We also learned that the energy to lift the bucket in Kitui comes from	The bucket gains 1600 J of PE when it is lifted 8 m, so this energy must come from somewhere, because energy cannot be made from nothing. The pulley cannot create energy. We still need to explain why the pulley makes the effort smaller and

	J for the 0.5 kg pendulum bob raised 0.2 m. With friction turned on in the simulation, the swing slowly died down.	the chemical energy in the person's body from food such as ugali and githeri, not from the pulley.	whether any energy is lost in a real system; we will investigate these questions in the next lessons.
Lesson 4 Conservation of Energy in Applications: Where Does the Bucket's Energy Go in a Real Fall?	We observed that the real bucket fall measured 11 m/s at the water, not the ideal 12.6 m/s. Its KE was 1210 J instead of 1600 J. A sliding block became warm and made sound and stopped on a rough surface.	We learned that in real systems the total energy is still conserved, but some energy is transferred to the surroundings as heat and sound through friction. 'Lost' energy is not destroyed. The 1600 J of PE became 1210 J of KE plus 390 J of heat, sound and air/rope friction.	We can explain that the real bucket is slower than the ideal one because 390 J of its energy was transferred to heat and sound, so energy is conserved overall. This suggests that in Mama Kavata's pulley system some energy will also be transferred as heat by friction, which we will measure later.
Lesson 5 Work: $W = Fd$ – What Does Mama Kavata's Lift Really Do?	We observed that a 2 kg load lifted 1 m needed about 20 N of force, giving $F \times d = 20 \text{ J}$, and that holding the load still or carrying it sideways involved no distance in the direction of the force. For the bucket in the Kitui well, 200 N x 2 m, 4 m, 6 m and 8 m gave 400 J, 800 J, 1200 J and 1600 J, matching the PE table.	We learned that work is done when a force moves an object in the direction of the force, $W = Fd$, measured in joules. Work is positive when the force and motion are in the same direction and negative when they are opposite. Mama Kavata does +1600 J lifting the bucket, gravity does -1600 J, and the net work is 0 J at steady speed. When lowering the bucket, the signs are reversed.	The direct lift needs 200 N over 8 m, so the woman does 1600 J of work, which becomes the 1600 J of PE gained by the bucket. The energy comes from the chemical energy in her body from food such as ugali and githeri. We still need to find out how the pulley system changes the force and distance and whether it changes the work, which we will investigate in Lesson 6.
Lesson 6 Work and Energy: Hand Lift versus Pulley System at the Kitui Well	We observed that lifting the 20 kg bucket 8 m by hand needs 200 N over 8 m (1600 J), while with the pulley system the effort is only 125 N but the rope is pulled 16 m, giving 2000 J of work input. In our model lift, the pulley effort was smaller than the weight but the string moved further.	We learned that work done on an object equals its change in energy, so the bucket gains 1600 J of PE either way. The pulley does not reduce the work needed; it trades a smaller force for a longer distance. The extra 400 J (20%) is transferred as heat (and a little sound) by friction, so energy is conserved.	The pulley system feels easier for Mwende because she pulls with 125 N instead of 200 N, but she pulls 16 m instead of 8 m. Her body supplies 2000 J from chemical energy in food such as ugali and githeri; 1600 J becomes PE of the bucket and 400 J becomes heat in the pulleys and rope, so no energy is destroyed.
Lesson 7 Work Against Friction and Power: Dragging the Maize Sack and Lifting Water	We observed that dragging a sack of maize across rough ground needs a larger friction force (150 N) than on smooth concrete (100 N), so more work is done over the same 10 m (1500 J versus 1000 J). We also saw that the same stair climb (1500 J) takes 6 s when walking and 3 s when running, and that the pulley system needs 2000 J input in 20 s (100 W) while hand lifting needs 1600 J in 20 s (80 W).	We learned that friction does negative work on a moving object, so a person must do work equal to friction x distance to keep it moving at steady speed, and that this energy is transferred as heat. We learned that power is the rate of doing work, $P = W/t$, measured in watts where $1 \text{ W} = 1 \text{ J/s}$, and that doing the same work faster needs more power.	The pulley system needs 2000 J of input although the bucket gains only 1600 J of PE because friction in the pulleys and rope transfers 400 J to heat, just as friction turns the work of dragging the maize sack into heat. No energy is destroyed. The pulley lets Mwende use a smaller 125 N force over a longer 16 m rope pull, and its useful output power of 80 W equals the hand lift power while its input power is 100 W.
Lesson 8 Simple Machines, MA, VR, Efficiency and the Final Model: Why the Pulley Makes the Bucket	We observed that the pulley system needs an effort of 125 N for a 200 N load, with the rope pulled 16 m while the bucket rises 8 m. We observed that the ramp needs 200 N for a 500 N sack over 4 m and the crowbar	We learned that $MA = \text{load/effort}$, $VR = \text{effort distance/load distance}$ and $\text{efficiency} = (MA/VR) \times 100\%$. The pulley system has MA 1.6, VR 2 and 80% efficiency; the ramp has 2.5, 4 and 62.5%; the crowbar has 3.2,	The bucket gains 1600 J of PE whether Mama Kavata lifts it by hand or Mwende uses the pulley system. The pulley lets Mwende pull 125 N instead of 200 N, but over 16 m instead of 8 m, so 2000 J of

Feel Light	needs 125 N for a 400 N load. In all three machines the work input was more than the useful work output.	4 and 80%. A single fixed pulley has $VR = 1$ and only changes the direction of the force. A machine never reduces the work needed; it trades a smaller force for a longer distance.	chemical energy from food (ugali, githeri) is put in. 1600 J becomes PE of the bucket and 400 J (20%) is transferred as heat by friction in the pulleys and rope. Energy is conserved because none is created or destroyed.
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